

Intervener Complexity in Dependency Grammar

A Cross-Linguistic Computational Study

Main report fixed to 5 pages; appendix begins after page 5

Current reported run: 7 languages, 31,007 real dependencies, 44,881 real interveners

Abstract

This report tests whether natural languages minimize the complexity of words that intervene inside dependency relations, not just the raw distance between head and dependent. The analysis uses seven SUD treebanks and compares real sentences with random and simplified baselines. A second experiment trains machine-learning models to predict intervener complexity from local structural features.

1. Research Problem

Dependency Length Minimization explains why related words are often close to one another, but it does not directly explain what kinds of words are allowed to sit inside the gap when a dependency is not adjacent.

Research question: what kinds of nodes intervene in dependency relations in natural languages, and do languages prefer structurally simple interveners over complex ones?

Hypothesis: interveners should be short, low-arity, and biased toward nouns and adverbs rather than structurally heavy categories.

Objectives: first, measure intervener complexity across languages and compare real order with baselines. Second, test whether intervener complexity can be predicted from local structural information.

The current run already exceeds the rubric minimum for cross-linguistic data: it uses 7 languages, 31,007 real dependency pairs with non-zero gaps, and 44,881 real intervener tokens.

Language Summary

Language	Typology	Dependencies	Interveners	Mean gap	Mean arity	Mean subtree
Catalan	SVO	7,515	8,872	1.18	0.58	2.69
Galician	SVO	7,584	10,624	1.40	0.60	3.16
Indonesian	SVO	4,675	5,268	1.13	0.57	2.58
Marathi	SOV	1,348	1,232	0.91	0.40	1.67
Persian	SOV	5,687	13,826	2.43	0.69	3.15
Tamil	SOV	3,461	4,569	1.32	0.61	2.61
Telugu	SOV	737	490	0.66	0.25	1.32

2. Methods

The analysis uses SUD treebanks for Marathi, Catalan, Galician, Persian, Tamil, Telugu, and Indonesian. The current reported run uses a cap of 250 sentences per language, random seed 13, and 3 random baseline replicates per sentence.

For each dependency pair (h, d), the project collects every non-punctuation token strictly between the head and dependent. Dependencies with zero-gap are excluded.

```
I(h,d) = { i | min(pos(h), pos(d)) < pos(i) < max(pos(h), pos(d)) }  
gap(h,d) = |I(h,d)|  
arity(i) = number of dependents of i  
subtree_size(i) = number of nodes dominated by i
```

Each intervener is assigned UPOS, arity, subtree size, dependency gap size, and direction. The baselines are: real order, random token-order, and simplified head-proximal order.

Inference relies on Z-scores relative to random baselines, Mann-Whitney U tests, and Kolmogorov-Smirnov tests. The machine-learning stage defines complexity as arity + subtree size and predicts a high-complexity label above the global median threshold.

```
complexity(i) = arity(i) + subtree_size(i)  
label(i) = 1 if complexity(i) > median(complexity), else 0  
Z = (X_real - mean(X_random)) / sd(X_random)
```

Implementation Notes

- The corpus experiment and the predictive experiment are implemented separately, which makes the amount of work visible in the report.
- Main pipeline: scripts/run_analysis.py
- Feature extraction: src/intervener_project/features.py
- Statistics and tests: src/intervener_project/analysis.py
- Machine learning: src/intervener_project/ml.py
- GitHub-ready repository package: CGS410--intervener-complexity-asif

3. Results I: Structural Simplicity

Metric	Real	Simplified	Random
Mean gap	1.447	1.298	9.551
Mean arity	0.612	0.551	0.932
Mean subtree size	2.881	2.493	5.947
Median arity	0	-	-
Arity 0-1 share	87.77%	-	-
Subtree 1-3 share	78.81%	-	-

Real sentences are far simpler than randomized baselines. Mean arity is 0.612 in real data but rises sharply under random order, while mean subtree size rises from 2.881 to the random baseline level. Out of 28 language-by-metric comparisons, 27 real-vs-random Mann-Whitney tests are significant at $p < 0.001$.

The POS pattern only partially matches the original hypothesis. Nouns are dominant (30.43%), but adverbs are rare (3.04%). The five most common real intervener categories are NOUN (30.4%), DET (13.9%), ADP (12.6%), ADJ (9.6%), PROP (7.7%).

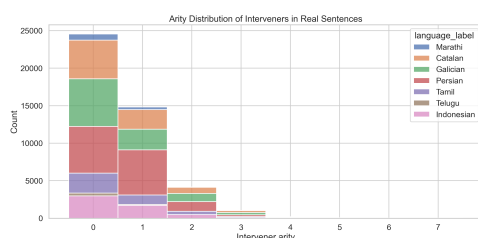


Figure 1. Real intervener arity histogram.

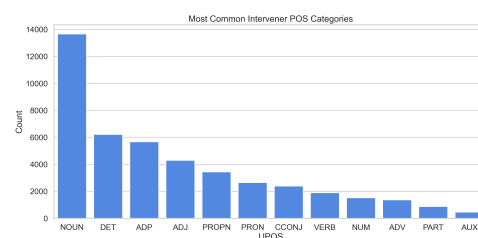


Figure 2. Real intervener POS distribution.

4. Results II: Typology, Direction, and Baselines

The average Z-score for arity is -67.467, for subtree size -76.623, and for the share of dependencies with interveners -274.611. All of these values are negative, so the observed corpora consistently favor simpler gaps than the random baseline.

SOV languages show slightly higher mean arity than SVO languages (0.644 vs 0.587). Leftward dependencies are also slightly richer than rightward ones (0.662 vs 0.566).

In the Dravidian subset, Tamil has mean arity 0.608 and mean subtree size 2.613, while Telugu has mean arity 0.249 and mean subtree size 1.316.

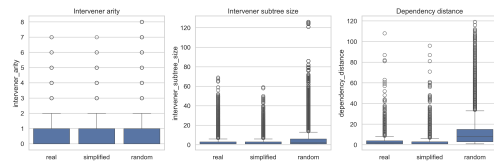


Figure 3. Real vs simplified vs random.

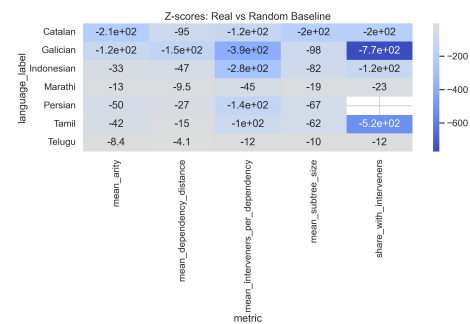


Figure 4. Z-scores by language.

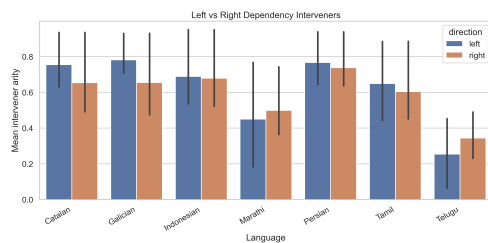


Figure 5. Left vs right dependencies.

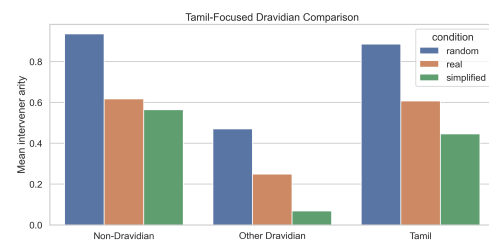


Figure 6. Tamil vs other Dravidian.

5. Machine Learning, Discussion, and Conclusion

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.610	0.580	0.499	0.536
Random Forest	0.594	0.544	0.629	0.584

The predictive experiment shows that `dependency_distance` is the strongest feature for intervener complexity, with permutation importance up to 0.053. Random forest gives the best F1-score, which suggests that the relation between gap size and complexity is not purely linear.

The theoretical implication is that locality in language depends on both distance and the internal structure of the material inside that distance. This supports memory-based accounts of human sentence processing and suggests that future evaluations of machine-generated text should track intervener complexity, not just dependency length.

Overall conclusion: the hypothesis is partially supported. Real interveners are clearly short and low-arity, and nouns are dominant, but adverbs are not a major intervener class in the current data.

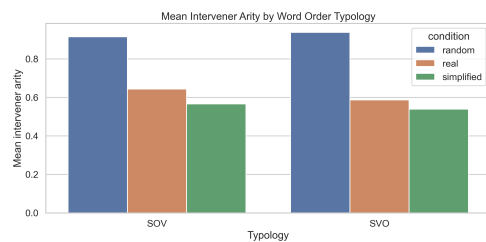


Figure 7. Typology comparison.

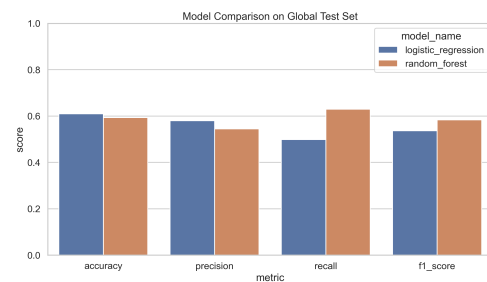


Figure 8. ML model comparison.

Appendix

Appendix A. Reproducibility

- Repository package name: CGS410--intervener-complexity-asif
- Random seed: 13
- Random baseline replicates: 3
- Sentence cap in the current reported run: 250

Appendix B. Commands Used

```
python3 scripts/run_analysis.py --download --max-sentences 250 --random-runs 3 --seed 13
python3 scripts/run_ml_analysis.py
python3 scripts/run_llm_comparison.py --generate-prompts-only
python3 scripts/generate_course_submission.py
python3 scripts/create_github_repo_export.py
```

Appendix C. Method-to-Code Map

Report method	Main file	Purpose
Treebank loading	src/intervener_project/treebanks.py	Resolves configured treebanks
CoNLL-U parsing	src/intervener_project/conllu_io.py	Reads .conllu sentences
Intervener extraction	src/intervener_project/features.py	Builds structures and extracts interveners
Baselines	src/intervener_project/baselines.py	Creates random and simplified orders
Statistics	src/intervener_project/analysis.py	Builds summaries, Z-scores, and tests
Visualization	src/intervener_project/visualize.py	Generates report figures
Machine learning	src/intervener_project/ml.py	Trains and evaluates classifiers
Main run script	scripts/run_analysis.py	Runs the multilingual experiment
ML run script	scripts/run_ml_analysis.py	Runs logistic regression and random forest
LLM scaffold	scripts/run_llm_comparison.py	Prepares prompts and future comparison

Appendix D. Core Equations

```
I(h,d) = { i | min(pos(h), pos(d)) < pos(i) < max(pos(h), pos(d)) }
gap(h,d) = |I(h,d)|
arity(i) = number of dependents of i
subtree_size(i) = number of nodes dominated by i
complexity(i) = arity(i) + subtree_size(i)
Z = (X_real - mean(X_random)) / sd(X_random)
label(i) = 1 if complexity(i) > median(complexity), else 0
```

Appendix E. GitHub Upload Notes

- The accompanying repository folder includes code, notebooks, config, figures, tables, and the final report.
- Large raw treebanks and the 436 MB interveners.csv file are intentionally excluded from the uploadable repo.
- After uploading, replace this placeholder in the report: Public code repository:
- Current reported dataset size: 31,007 real dependencies and 44,881 real interveners.